Sinothando Mafu 25 October 2026

SQL Exercise 4

UNION & UNION ALL

Q1:

```
SELECT account_id, account_holder, city
FROM branch_sandton_accounts
UNION
SELECT account_id, account_holder, city
FROM branch_rosebank_accounts;
```

R:

| account_id | account_holder | city |
|---|---|---|
| A001 | Nomvula Dlamini | Johannesburg |
| A002 | David Makoena | Johannesburg |
| A003 | Lerato Sithole | Johannesburg |
| A004 | Peter Nkosi | Johannesburg |
| A005 | Zanele Khumalo | Johannesburg |
| A006 | Thabo Motha | Johannesburg |

Q2:

```
SELECT product_code, product_name, product_type
FROM savings_products
UNION
SELECT product_code, product_name, product_type
FROM current products;
```

R:

| product_code | product_name | product_type |
|---|---|---|
| SV01 | Basic savings | savings |

| | | |
|---|---|---|
| SV02 | Premium Savings | Savings |
| SV03 | Youth Savings | savings |
| SV04 | Business Savings | Savings |
| CR01 | Standard Current | Current |
| CR02 | Gold current | Current |
| CR03 | Business Current | Current |

Q3

```
SELECT staff_id, staff_name, email
FROM retail_banking_staff
UNION
SELECT staff_id, staff_name, email
FROM corporate_banking_staff
WHERE staff_id NOT IN(SELECT staff_id FROM
retail_banking_staff)
ORDER BY staff_id;
```

R:

| Staff_id | staff_name | email |
|---|---|---|
| S001 | Mpho Radebe | mpho01.radebe@nexbank.co.za |
| S002 | Brian Tshabalala | briant23@nexbank.co.za |
| S003 | Aisha Patel | paisha09@nexbank.co.za |
| S004 | Kabelo Moabelo | kabelo.moabelo@nexbank.co.za |
| S005 | Nandi Dube | nandi.d@nexbank.co.za |
| S006 | Sipho Khumalo | sipho.k@nexbank.co.za |

Q4:

```
SELECT city_code, city_name, region
FROM mobile_branch_cities
UNION
SELECT city_code, city_name, region
```

FROM digital_branch_cities;

R:

| city_code | city_name | region |
|---|---|---|
| CO1 | Johannesburg | Gauteng |
| CO2 | Pretoria | Gauteng |
| CO3 | Cape Town | Western Cape |
| CO4 | Durban | KwaZulu - Natal |
| CO5 | Polokwane | Limpopo |
| CO6 | Port Elizabeth | Eastern cape |

Q5:

SELECT customer_id, customer_name, segment
FROM push_notification_targets
UNION
SELECT customer_id, customer_name, segment
FROM inapp_banner_targets;

R:

| customer_id | customer_name | segment |
|---|---|---|
| C1001 | Nomsa Zwane | Retail |
| C1002 | Andile Buthelezi | Retail |
| C1003 | Fatima Mahomed | SME |
| C1004 | Ryno van Zyl | corporate |
| C1005 | Thandeka Cele | Retail |
| C1006 | Samuel Nkosi | Private Wealth |

Q6:

```
SELECT transaction_id, account_id, amount, transaction_date
FROM atm01_transactions
UNION ALL
SELECT transaction_id, account_id, amount, transaction_date
FROM atm02_transactions;
```

R:

| transaction_id | account_id | amount | transaction_date |
|---|---|---|---|
| T1001 | A001 | 500.00 | 2026-03-01 |
| T1002 | A002 | 1200.00 | 2026-03-01 |
| T1003 | A001 | 300.00 | 2026-03-02 |
| T1004 | A003 | 750.00 | 2026-03-02 |
| T1003 | A001 | 300.00 | 2026-03-02 |
| T1005 | A004 | 900.00 | 2026-03-03 |
| T1006 | A002 | 450.00 | 2026-03-03 |
| T1007 | A005 | 150.00 | 2026-03-04 |

Q7:

```
SELECT application_id, customer_id, loan_type, amount_requested
FROM gauteng_loan_applications
UNION ALL
SELECT application_id, customer_id, loan_type, amount_requested
FROM western_cape_loan_applications;
```

R:

| application_id | customer_id | loan_type | amount_requested |
|---|---|---|---|
| LA001 | C1001 | Home Loan | 850000.00 |
| LA002 | C1002 | Vehicle Loan | 220000.00 |
| LA003 | C1003 | personal | 45000.00 |
| LA004 | C1004 | Home Loan | 1150000.00 |

| | | | |
|---|---|---|---|
| LA003 | C1003 | personal | 45000.00 |
| LA005 | C1005 | Vehicle Loan | 310000.00 |
| LA006 | C1006 | Business | 550000.00 |
| LA007 | C1007 | Home Loan | 920000.00 |

Q8:

```
SELECT complaint_id, customer_id, category, logged_date
FROM email_complaints
UNION ALL
SELECT complaint_id, customer_id, category, logged_date
FROM app_complaints;
```

R:

| complaint_id | customer_id | category | logged_date |
|---|---|---|---|
| EC001 | C2001 | Card Dispute | 2026-03-05 |
| EC002 | C2002 | Online Banking | 2026-03-06 |
| EC003 | C2003 | ATM Failure | 2026-03-07 |
| EC004 | C2004 | Account Fees | 2026-03-08 |
| AC001 | C2005 | App Crash | 2026-03-05 |
| AC002 | C2001 | Card Dispute | 2026-03-06 |
| AC003 | C2006 | Payment Delay | 2026-03-07 |
| AC004 | C2007 | Login Issue | 2026-03-08 |

Q9:

```
SELECT payment_id, account_id, amount, payment_date
FROM april_payments
UNION ALL
SELECT payment_id, account_id, amount, payment_date
FROM may_payments;
```

| payment_id | account_id | amount | payment_date |
|---|---|---|---|
| PAY001 | A001 | 12500.00 | 2026-04-15 |
| PAY002 | A002 | 4800.00 | 2026-04-18 |
| PAY003 | A003 | 9200.00 | 2026-04-25 |
| PAY004 | A004 | 3300.00 | 2026-04-28 |
| PAY005 | A001 | 12500.00 | 2026-05-15 |
| PAY006 | A005 | 7600.00 | 2026-05-20 |
| PAY007 | A002 | 5100.00 | 2026-05-22 |
| PAY008 | A006 | 2800.00 | 2026-05-27 |

Q10:

```
SELECT entry_id, account_id, entry_type, amount, entry_date
FROM debit_entries
UNION ALL
SELECT entry_id, account_id, entry_type, amount, entry_date
FROM credit_entries;
```

R:

| entry_id | account_id | entry_type | amount | entry_date |
|---|---|---|---|---|
| DR001 | A001 | Debit | 1500.00 | 2026-03-10 |
| DR002 | A002 | Debit | 3400.00 | 2026-03-11 |
| DR003 | A003 | Debit | 850.00 | 2026-03-12 |
| DR004 | A004 | Debit | 2100.00 | 2026-03-13 |
| CR001 | A001 | Credit | 5000.00 | 2026-03-10 |
| CR002 | A005 | Credit | 1200.00 | 2026-03-11 |
| CR003 | A002 | credit | 4500.00 | 2026-03-12 |
| CR004 | A006 | credit | 3100.00 | 2026-03-13 |

## Bonus Questions

Q1:

UNION,

The analyst wants a unique list, meaning distinct customers and the removal of duplicate rows. A customer who appeared on both the January and February watch lists would show up only once. UNION ALL would list them twice, which is not what the analyst wants.

Q2:

UNION ALL

Every transaction is retrieved, including duplicates. This keeps every row, which is what the auditing team asked for.

Q3:

The column city was not included in the first query, it returns only the account_id and account_holder columns whereas the second query returns the aforementioned columns including city. Both SELECT statements in a UNION must have the same number of columns in the same order, with compatible/matching data types, so the query fails.